KAREL KUTCHAN

Software Engineer and Technical Analyst | Requirements analysis | API and integration design

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PROFESSIONAL SUMMARY

Software engineer with more than ten years of commercial experience delivering web and enterprise applications. Since February 2026 I design and build Seeder, a production system for the National Library of the Czech Republic on Django and PostgreSQL, where curators come to me with a need and I turn it into a data model, workflows, implementation and deployment. Until April 2026 I led frontend engineering at BlueGhost, owning architecture, delivery standards and quality across product teams.

I do my best work at the boundary between business and IT. I ask until the real need is clear, structure the problem, propose solution options and turn them into a specification a team can build from. My background includes the Moje MND customer portal, internal process applications for E.ON and three years selling CRM/ERP solutions, so I know both sides of a requirement: how it sounds from the customer and what has to come out of it.

I work with REST interfaces and typed API contracts, PostgreSQL data models and SQL, Git, pull requests and CI/CD. Documentation, decision records and specifications are part of my delivery, not an afterthought. I use AI tooling (GitHub Copilot, agentic CLI workflows) in a structured way and keep its output under code review. I continue to study as a core student at 42 Prague.

CORE SKILLS

Analysis and design	Requirements gathering and validation, problem structuring, solution options, functional specification, data modelling (PostgreSQL), REST contract design, technical documentation and decision records
Integration and backend	REST APIs, Python and Django, PostgreSQL and SQL, Node.js scripts and tooling, CMS and headless platform integration (Payload CMS, Kontent.ai, Kentico), PHP/Twig
Frontend engineering	React 18/19, TypeScript, Next.js, JavaScript, HTML5, CSS3, design systems, Storybook
Data and forms	TanStack React Query, Zustand, Redux, typed API integration, configuration-driven UI, schema-aware validation
Quality and delivery	Git, GitHub, pull requests, code review, CI/CD (Jenkins), Docker, Playwright, Vitest, React Testing Library, mentoring
AI-assisted work	GitHub Copilot, agentic CLI workflows (Claude Code, Codex), system prompts and agent rules, Figma-to-token and code assistance, AI output always under code review and tests
Accessibility and UX	WCAG-oriented implementation, React Aria, collaboration with UX, internationalisation-aware UI

EXPERIENCE

Software Engineer | Národní knihovna ČR | 02/2026 – Present

Stack: Python, Django, PostgreSQL, Docker, Playwright, Git

- Seeder is a production Django application on PostgreSQL in which Web Archive curators register web sources and publishers, track licensing agreements, schedule harvests and assemble thematic collections for webarchiv.cz.
- I analyse curators' needs, propose changes to workflows and the data model, and take them from specification through implementation to deployment, working across the database, application logic, admin and UI.
- Changes ship through pull requests, Playwright tests and Docker, deployed with Jenkins to the test and production environments.

Lead Frontend Engineer | BlueGhost | 03/2025 – 04/2026

Stack: Next.js, React, TypeScript, PNPM, TanStack React Query, Zustand, SCSS Modules, Storybook, React Aria, Vitest, Playwright, Git

- Owned frontend architecture and delivery standards across product teams, and documented solution designs and technical decisions so the team could take them over and extend them.
- Refined requirements with stakeholders and backend colleagues before development started, and standardised server-state and client-state handling (TanStack React Query, Zustand) so data flows and cache invalidation were easy to reason about.
- Improved release confidence by combining Storybook with Vitest and Playwright testing and stricter code review instead of relying on manual QA.
- Introduced structured AI-assisted development practices (GitHub Copilot, CLI workflows, system prompts) while keeping architecture, code review and maintainability under engineering control.

Frontend Engineer | Kontent.ai | 06/2024 – 02/2025

Stack: React, TypeScript, Next.js, styled-components, React Aria, Storybook, Playwright, Git

- Analysed the accessibility of a complex React application against strict requirements (landmarks, labels, focus behaviour, component semantics), then designed and carried out a refactoring of interaction patterns with React Aria.
- Worked in a contract engagement with a defined delivery scope, on maintainable TypeScript components, Storybook and Playwright checks.

Lead Frontend Engineer | TLDR;IT s.r.o. | 04/2024 – 09/2024

Stack: Next.js, React, TypeScript, Sass, CSS Modules, Payload CMS

- Designed the frontend architecture for time-boxed web and mobile-oriented projects and connected React interfaces to Payload CMS content through TypeScript models and clearly defined data contracts.

Frontend Developer | eMan a.s. | 09/2021 – 04/2024

Stack: React, Next.js, TypeScript, MobX, Redux, Sass, CSS Modules, Storybook, Git

- Built the Moje MND customer portal from scratch in a two-person frontend team: with business and backend we refined the requirements for contracts, invoices, forms and document handling and turned them into screens, validation and API contracts.
- Delivered two internal process applications for E.ON: PUDplus for work orders, findings and preventive maintenance, and MKME web for work orders and metering.
- Worked for E.ON, MND, ČEZ and Bohemia Energy in close collaboration with UX and backend teams on production applications with large user bases, increasing reuse through design systems and component libraries.

Frontend Developer | LMC | 09/2020 – 09/2021

Stack: React, JavaScript, PHP/Twig, Less, Sass

- Translated corporate clients' and designers' requirements into custom web solutions on the jobs.cz ecosystem, in a hybrid PHP/Twig and frontend environment.

Frontend Developer | Amp X | 09/2019 – 09/2020

Stack: React, TypeScript, Redux, JSS, Storybook, Git

- Built React and TypeScript interfaces for a distributed-energy platform where data clarity and reliable interaction patterns were critical for market-facing processes.

Frontend Developer | Skype.com | 04/2019 – 09/2019

Stack: JavaScript, jQuery, HTML5, CSS3, Sass, Kentico CMS, Git

- Contributed to a public website serving more than 400,000 unique users daily in more than 30 countries, aligning frontend components with editorial workflows in a Kentico publishing environment.

Frontend Coder | Foxconn DRC s.r.o. | 06/2018 – 03/2019

Stack: HTML5, CSS3, Sass, React, BEM, Git

- Built scalable UI foundations for data visualisation, management and processing applications on a BEM methodology.

Frontend Coder | Mountfield, a.s. | 05/2017 – 05/2018

Stack: HTML5, CSS3, JavaScript, jQuery, Kentico CMS

- Maintained the main company website and marketing microsites in a Kentico CMS-driven environment.

Web Administrator, Graphic Designer, and Junior Frontend Coder | BitWare CZ s.r.o. | 08/2014 – 05/2017

Stack: HTML5, CSS3, JavaScript, jQuery, PHP, WordPress

- WordPress site administration, visual design and first frontend delivery in HTML, CSS, JavaScript and PHP.

Earlier experience: Sales Manager · Austro-Bohemia (2011–2014) — selling and rolling out CRM/ERP and ICT services, gathering customer requirements and translating between business and supplier.

EDUCATION & LANGUAGES

42 Prague (2026–Present)
Core student — software engineering fundamentals in C and Python

COPTH (2002–2006)
Electrical engineering, telecommunications and computer technologies

Czech — Native
English — Working professional
Swedish — Beginner